

D12 — Contrôle double-mode PAC / Arrosage

Dual-Mode Control — Heat-pump (PAC) / Irrigation with automatic detection

This document describes the `dual-mode-control.yaml` ESPHome package, which turns the D12 + D1 mini into an autonomous variable-speed pump controller serving two uses from the same 100 L buffer tank:

- PAC mode — the ground-source heat pump draws well water through a low-resistance exchanger. Goal: lowest energy (≈ 144 W) at a low fixed economy frequency.
- Irrigation mode — sprinklers/nozzles, a high-resistance circuit. Goal: regulated pressure (≈ 2.5 bar) with automatic sleep when taps are closed.

The two uses are told apart automatically, from the hydraulics alone — no extra wire to the heat pump.

Do not use the heat-pump “demand” signal as a direct pump-start command: on this installation that same signal also drives the exchanger valve during the heat pump’s power-on purge, so it is not a clean indication of a real heating draw. The restored external start condition is the pressostat low-threshold contact; PAC/irrigation classification remains hydraulic.

The package is optional and self-contained. It is loaded from the main config via `packages: dual_mode:`
`!include dual-mode-control.yaml`. Comment that line to disable it.

1. Principle of automatic detection

Both consumers draw a similar flow (PAC ≈ 2.44 m³/h, irrigation ≈ 2.0 m³/h), so the rate of pressure drop does not distinguish them. What differs by $\sim 6\times$ is the hydraulic resistance of the active circuit:

	Circuit	Resistance	Pressure at a fixed frequency
PAC	exchanger, large bore	low (“soft”)	low ($\sim 0.4\text{--}1$ bar)
Irrigation	nozzles	high (“stiff”)	high (≈ 2.5 bar)

So the controller imposes a reference probe frequency (F_{probe} , ~ 38 Hz) for a few seconds at start-up and reads the resulting operating point (steady pressure + current):

- low pressure + real current $\Rightarrow$ PAC $\rightarrow$ economy fixed-frequency mode;
- high pressure $\Rightarrow$ irrigation $\rightarrow$ PID pressure regulation to the setpoint;
- high pressure + negligible current $\Rightarrow$ no draw $\rightarrow$ go back to sleep.

The buffer tank only adds a short settling delay; at equilibrium it does not hide the consumer’s signature.

2. Architecture

- Brain = ESPHome (P0.02 = Communication, P0.03 = Communication): start/stop, mode detection, frequency command, PID, sleep. All logic is visible and tunable from Home Assistant.

- External demand = pressure contact, not PAC demand: the signal exposed to the pump side is the pressostat low-threshold contact. The raw heat-pump demand contact is not used as a direct start input because it also actuates the exchanger valve during purge.
- Independent hardware safety net = the mechanical pressure switch wired to the D12 external-fault input, set at wide limits (0.1 / 5.0 bar) — dry-run and over-pressure protection that works even if the ESP fails. Plus the D12's own over-current / over-temperature protections.

This is the design chosen for this installation: a flexible software brain plus a simple hardware guardian.

3. State machine

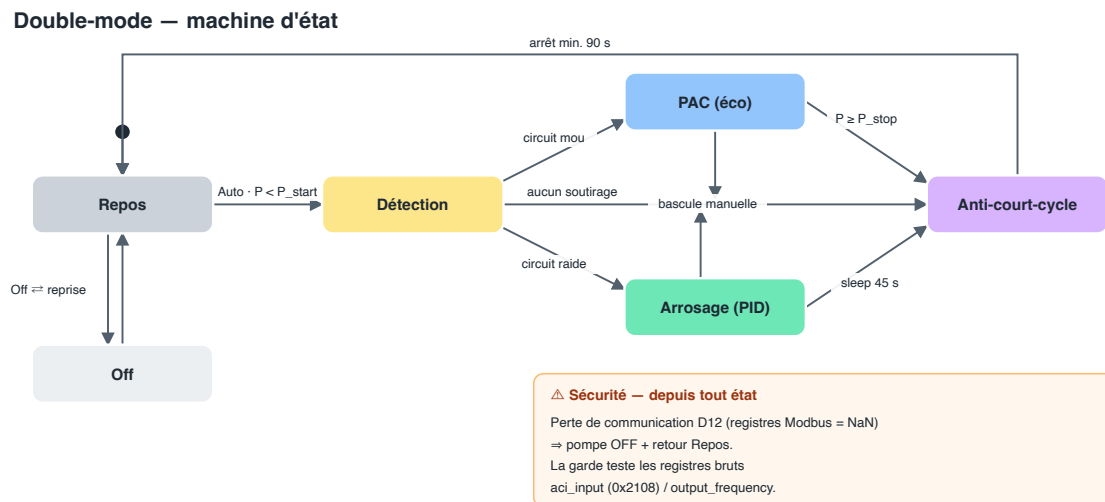


Figure 1: Dual-mode state machine

State names map to the DM État text sensor: Repos (IDLE), Détection (PROBE), PAC (éco), Arrosage (PID), Anti-court-cycle (COOLDOWN).

Mode selector (DM Mode)

Option	Behaviour
Auto	full automatic detection (probe → PAC or Irrigation)
PAC	force economy fixed-frequency mode (skip probe)
Arrosage	force PID pressure regulation (skip probe)
Off	controller idle, pump kept stopped — default after a fresh flash

4. Tunable parameters (Home Assistant)

Entity	Default	Meaning
DM P start (marche)	1.5 bar	pressure below which a cycle starts
DM P stop (arrêt PAC)	2.0 bar	PAC fill/stop target (draw-ended detection)

Entity	Default	Meaning
DM P consigne arrosage	2.5 bar	irrigation PID setpoint
DM F éco PAC	38 Hz	fixed economy frequency in PAC mode
DM F sonde (détection)	38 Hz	reference frequency during detection
DM P seuil classification (Ã calibrer)	1.4 bar	calibrate — PAC below / irrigation above, at F probe
DM I mini débit (Ã calibrer)	0.3 A	calibrate — min current proving a real draw
DM F max	50 Hz	upper frequency limit
DM F min (plancher sleep)	22 Hz	lower frequency / sleep floor
DM PID Kp (Hz par bar)	8.0	irrigation PID proportional gain
DM PID Ki (Hz par bar.s)	2.0	irrigation PID integral gain

Fixed timers (in the lambda): probe 10 s, min run 60 s, min off 90 s, sleep delay 45 s, max run 20 min (runaway guard).

The 11 parameters carry `entity_category`: `config` (grouped under Configuration on the device page); DM État / DM Mode détecté are diagnostic. A ready-made dashboard with an explicit description of every parameter — Home Assistant has no native hover tooltip — is provided in [lovelace-dual-mode.yaml](#) (legend card + adjustable controls, no add-on required).

5. Use-case timing diagrams

Case A — PAC cycle (Auto)

Detection finds a soft circuit, the pump runs at the economy frequency, fills the tank, stops at `P stop`, then waits out the anti-short-cycle delay.

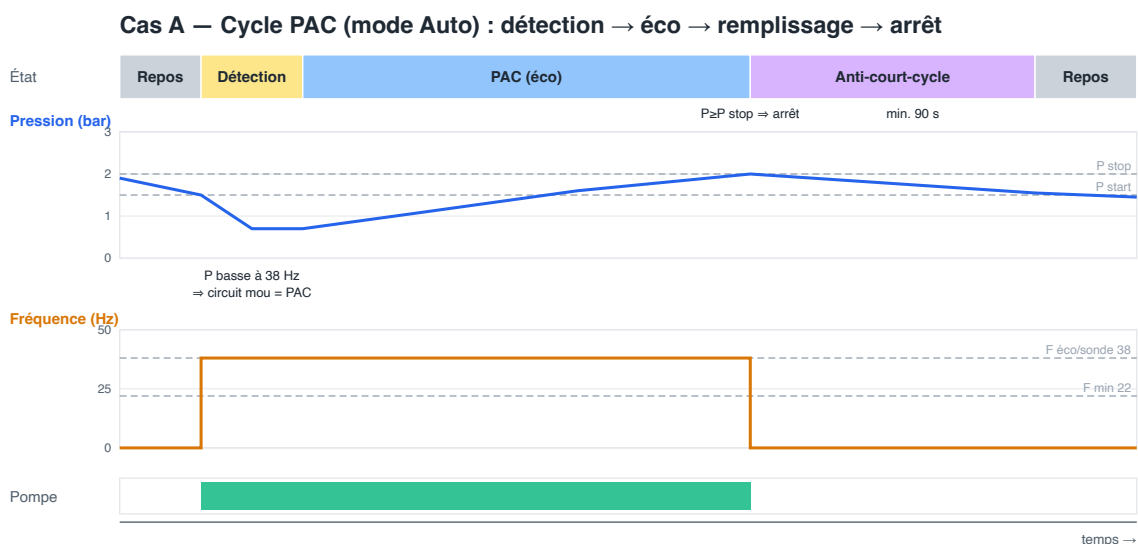


Figure 2: PAC cycle

Case B — Irrigation with sleep / wake

Detection finds a stiff circuit, the PID holds 2.5 bar; when the tap closes the frequency falls to $F_{\min}$, and after 45 s the pump sleeps. Reopening a tap wakes it.

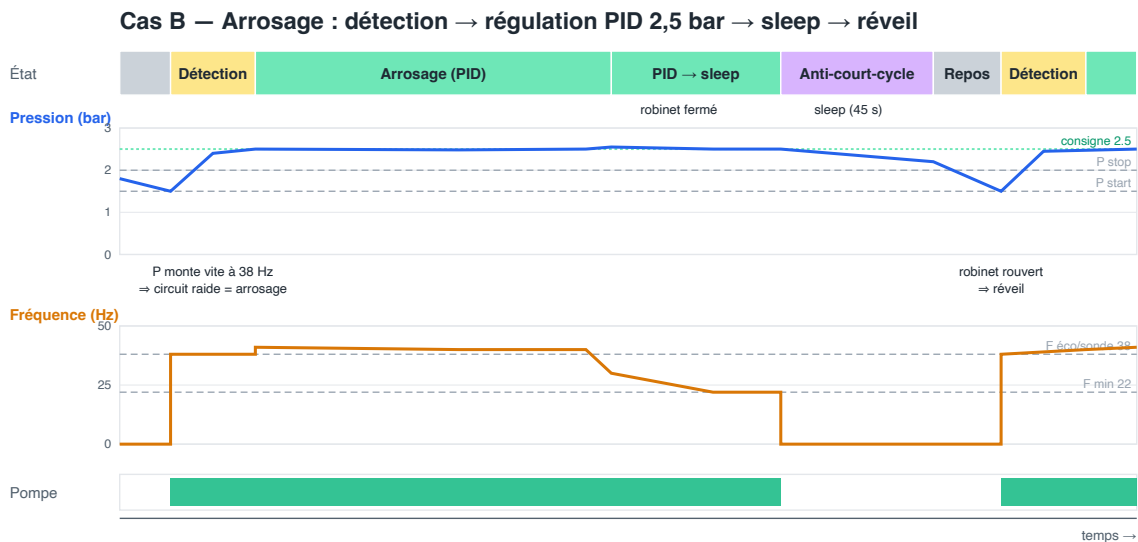


Figure 3: Irrigation + sleep

Case C — PAC purge ride-through

At every power-up the heat pump runs a ~3 min air purge with intermittent short draws. Because start/stop is driven by tank pressure / the pressostat low-threshold condition rather than by the raw heat-pump demand signal, the exchanger-valve purge does not become a direct pump-start command. The anti-short-cycle timer caps restarts, and the saw-tooth purge is absorbed by the buffer tank.

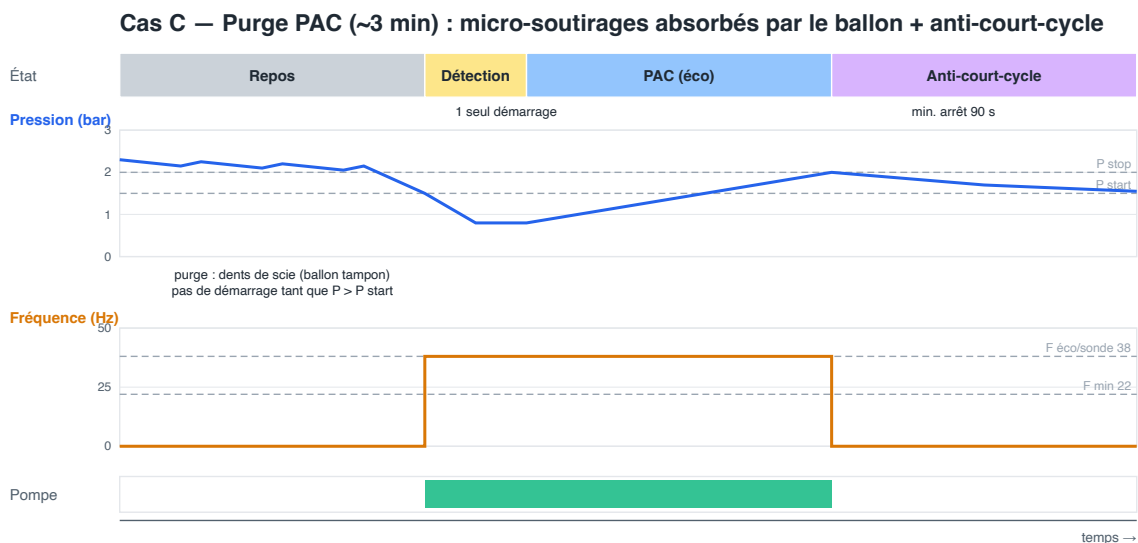


Figure 4: Purge handling

Case D — Communication-loss safety

If the D12 stops answering, the raw Modbus registers read NaN; the “comms-alive” guard forces the pump OFF and the state back to Repos. The guard checks the raw registers `aci_input` (0x2108) / `output_frequency`, not the derived Pressure sensor.

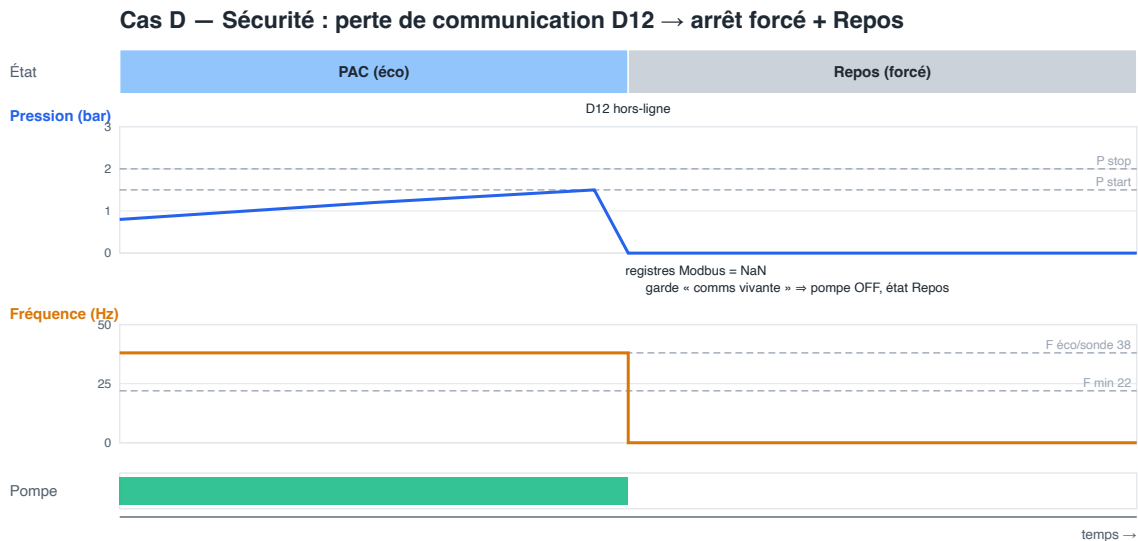


Figure 5: Comms loss

6. Commissioning & calibration

Do this in order:

1. Hardware safety first — wire the mechanical pressure switch (0.1 / 5.0 bar) to the D12 external-fault input. Do not run automatically before this is in place.
2. Pressure — wire the 4-20 mA sensor to ACI/GND and set the AI jumper to current (Cin); ESPHome reads it directly from register 0x2108 (raw ACI), no PID/analog scaling on the D12 needed.
3. D12 parameters — `P6.01` = 0000 (9600, the only supported rate), `P6.00` = 1, then for ESPHome control `P0.02` = 2 and `P0.03` = 6.
4. Calibrate detection — run the pump at `F probe` (38 Hz) once in PAC only and once in irrigation only; read the steady pressure each time. Set `DM P seuil classification` halfway between them. Read the running current to set `DM I mini débit`.
5. Go live — keep `DM Mode` = Off, then switch to Auto; watch `DM État` and `DM Mode détecté`. Tune `Kp` / `Ki` if the irrigation pressure oscillates.

7. Safety summary

- Default mode Off on a fresh device — never auto-runs the pump until explicitly set to Auto.
- Comms-alive guard — no pump command unless the D12 is answering.
- Anti-short-cycle (min on/off) and max-run software guards.
- Mechanical pressure switch at wide limits = independent hardware protection.
- These software guards do not replace the hardware pressure switch.

Wiring — ESPHome link (D1 mini ↔ RS485 ↔ D12) and the safety pressure switch (NC contact → D12 external-fault input X2, `P2.14` = 9):

Câblage : carte ESPHome (D1 mini ↔ RS485) + pressostat de sécurité → D12

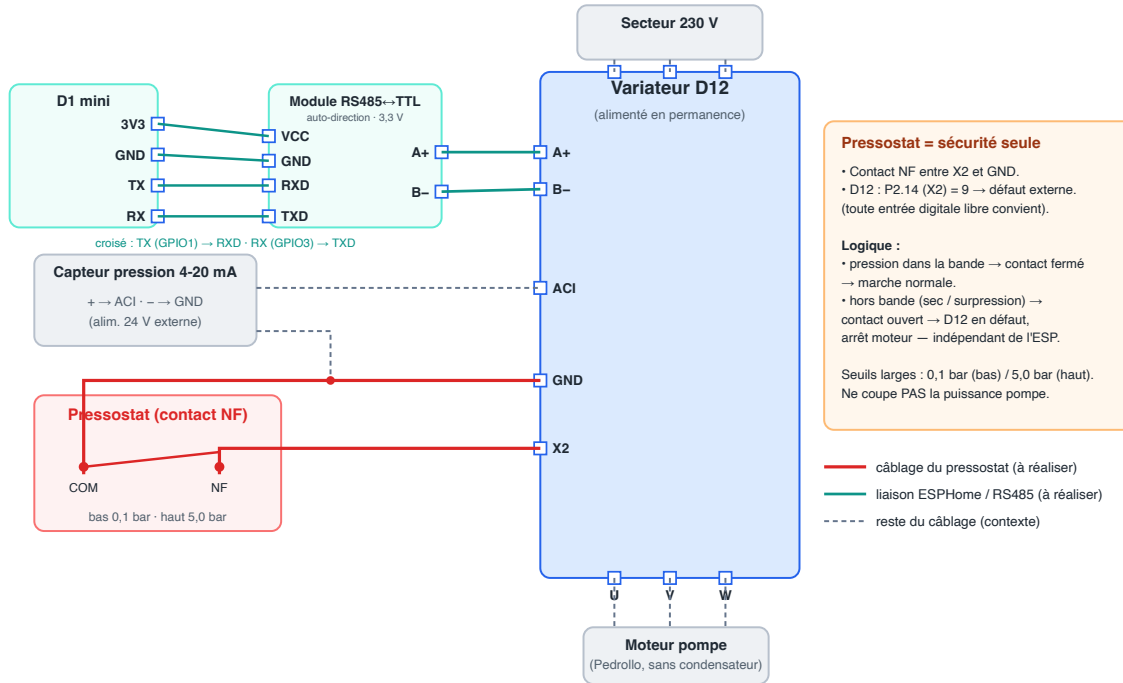


Figure 6: ESPHome and pressure-switch wiring

8. Wiring note (RS485 module)

This installation uses an auto-direction RS485↔TTL module (MAX485 + direction-control IC, 3.3 V / 5 V compatible). There is no DE/RE pin to wire, so ESPHome's `flow_control_pin`: GPIO5 is unused and can be removed (freeing GPIO5). Power the module at 3.3 V so logic levels match the D1 mini. See the main [README](#) for the full pinout.

If the fitted module is instead a manual MAX485-style board with DE and /RE pins, tie DE and /RE together, wire them to one ESP GPIO, and configure ESPHome `flow_control_pin`. Without that direction control, the module can transmit requests but stay unable to receive replies.

9. D12 parameter checklist

Do not trust the factory defaults — not all of them match what the integration expects. On this unit, P6.01 was found at 0000 (9600 bps), not the documented 0001 (19200), which broke all Modbus communication. Verify every parameter below. ! marks values that must differ from the factory default.

9.1 Communication (required to talk at all)

Param	Required	Default	Note
P6.00 local address	1	1	must equal modbus_address (1)

Param	Required	Default	Note
P6.01 comm config	0000	0001	! ZT-D12-220V is 9600-only — ones digit 1/2 are reserved, not 19200/38400. 0000 = 9600, no parity, normal response. ESPHome baud_rate must be 9600 too. Power-cycle the D12 after a baud change. ok (0 disables)
P6.02 timeout	10.0 s	10.0 s	

Digit map of P6.01 = [thousands] [hundreds] [tens] [ones]: ones = baud (0=9600; 1,2 reserved), tens = parity (0=none), hundreds = response (0=normal, 2=no reply).

9.2 Frequency scaling (so commanded Hz is correct)

Param	Required	Note
P0.04 max frequency	50.0 Hz	ESPHome sends the setpoint as % of P0.04; must match DM_F_{max} (50) or every Hz is wrong. Verify consistency
P0.05 upper limit	50.0 Hz	

9.3 Pressure reading — register 0x2108 (raw ACI)

The ZT-D12-220V exposes the raw analog inputs over Modbus, so ESPHome reads the 4-20 mA sensor directly from 0x2108 (ACI, value in mA × 100) — no PID feedback channel and no analog scaling on the D12 are required.

Item	Required	Note
AI jumper J5	Cin (current)	hardware — required for the 4-20 mA loop
4-20 mA sensor	wired to ACI / GND	+ external 24 V supply

ESPHome does the scaling $((mA - 4) / 16 \times range_bar)$, default 10 bar full scale). P3.00, P2.04–P2.07 and P3.18 are no longer needed for the reading (they only matter if you use the D12's own internal PID). This also removes the old “does 0x210E update in Communication mode?” unknown.

9.4 Control source (set before switching DM Mode to Auto)

Param	Required	Default	Note
P0.02 run command source	2 = Communication	0 = panel	! required for ESPHome to start/stop
P0.03 frequency source	6 = Communication	—	! required for ESPHome to set frequency

Param	Required	Default	Note
P0.09 thousands digit	0 = no PID overlay	0	The newer manual says thousands 1 means P0.03 + PID; leave it disabled because ESPHome is doing the pressure PID and writes the frequency over Modbus.

9.5 Safety pressure switch (only when X2 is wired)

Param	Required	Note
P2.14 (X2)	9 = external fault	! set only after the NC pressure switch is wired to X2, otherwise a floating X2 = permanent fault